

A Tiny AI WIRELESS STREAMING CAMERA

ASHWINI KUMAR SINHA

AI cameras are everywhere—spotting people, detecting objects, and powering advanced driver assistance systems in cars. But most are too bulky and power-hungry for compact applications like drones, wearables, or wildlife tracking.

This device demonstrates how machine learning can be embedded into a tiny package—perfect for places where size, power, and real-time intelligence matter. By using TensorFlow.js with the COCO-SSD model, the camera identifies people, vehicles, and animals directly within a browser, combining intelligent software with minimal hardware. This serves as the foundation for designing one of the smallest AI streaming cameras. Fig. 1 shows the author detecting a person and a potted plant during testing.

Why go small?

Compact AI cameras offer clear advantages across diverse sectors. In drones, they lower payload while supporting obstacle detection and object tracking. In wearables, they integrate discreetly into glasses or badges to deliver real-time visual feedback. In home security, they enable unobtrusive surveillance. For wildlife monitoring, their size allows placement in tight or concealed spots such as birdhouses. In robotics, they serve as visual inputs for navigating constrained environments.

These use cases

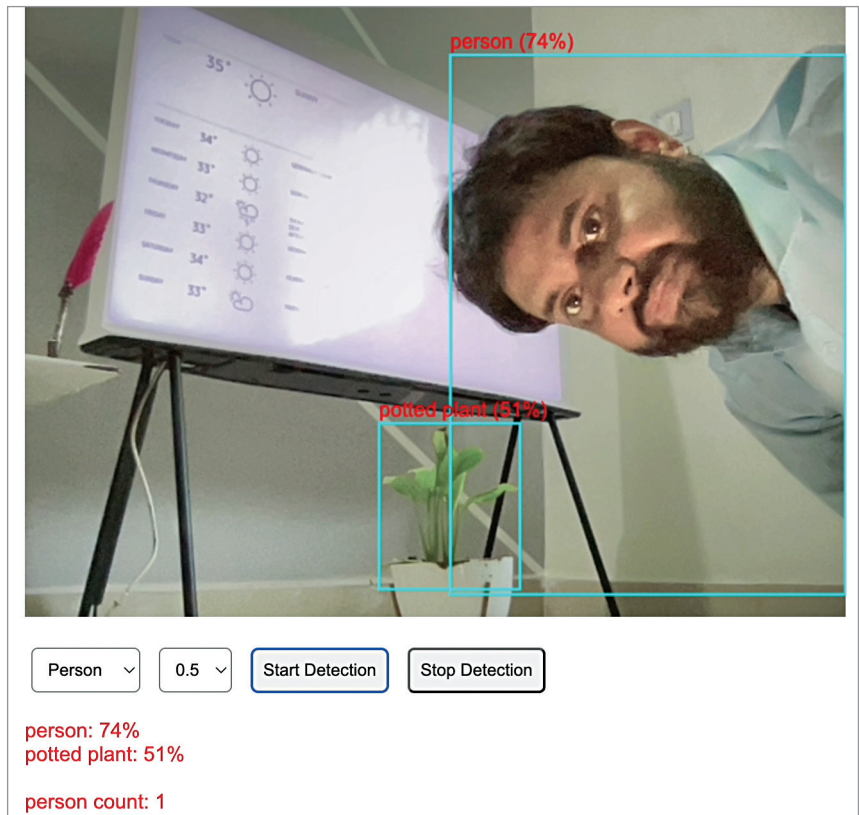


Fig. 1: Author detecting a person and a potted plant during testing

require devices that are intelligent, low-power, and exceptionally compact. A development board meeting all three criteria—capable of capturing frames, streaming video, and maintaining a minimal

BILL OF MATERIALS

Item	Remarks	Quantity
IndusBoard Coin V2	Tiny board	1
OV2640 camera module	2MP camera with JPEG output	1
Jumper wires	For hooking it all up	15
Power supply	3.3V battery or USB adaptor	1
Micro USB cable	For programming and power	1

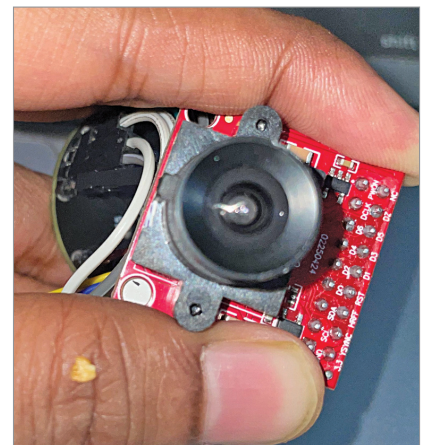


Fig. 2: Author's prototype